AutoTerm-SST: Adaptive Terminology Memory with Zero Session-Time Setup for Streaming Speech Translation

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Abstract

We present AutoTerm-SST, an interactive system for terminology-aware streaming speech translation. Live domain talks contain rare entities and technical terms, yet requiring users to prepare a glossary or select a domain before a session creates substantial friction. AutoTerm-SST provides zero session-time terminology setup by routing among domain slices registered offline: each streaming chunk injects at most 10 score-filtered glossary references into the prompt, normally drawn from a compact active slice, and a training-free router selects the active domain. The router combines generated target-translation topic evidence, speech-window domain probes, speech-centroid evidence, and retrieved-term metadata, without running a separate ASR or source-transcript classifier in the production path. The demo exposes retrieved terms, active slices, router decisions, and retrieval latency in a live evidence panel. On a ten-talk alternating ACL/medicine stream scored against forced-aligned term timings, automatic mode reaches 0.874 combined term accuracy, compared with 0.679 and 0.805 for fixed NLP and medicine inventories, and exceeds each fixed inventory inside its own domain. Scale probes show warm MaxSim retrieval stays near 60–105 ms (p50–p95) for memories of up to one million terms. Across the ten-talk stream, the router reaches the correct glossary after all nine domain transitions, with one transient, self-correcting excursion. The released implementation supports both mock and live-GPU modes.

1 Introduction

Simultaneous speech translation (SST) targets technical talks, lectures, meetings, and live media. Prior SST work studies the quality–latency trade-off of prefix translation (Ma et al., 2019) and standardizes streaming evaluation (Ma et al., 2020). Recent omni speech-language models broaden the available modeling options for long-form speech translation (Qwen Team, 2025), but live systems still struggle with term-dense content: named methods, datasets, clinical terms, legal phrases, and financial entities often require specialized translations that are difficult to infer from local acoustic context alone.

Glossaries are a practical way to control terminology, but they are awkward for live use. A user may not know the exact topic before a talk starts, and asking every speaker or viewer to upload a domain glossary or select a domain creates friction. At deployment time, operators register modular, pre-indexed terminology resources for different domains. At session time, users neither upload a glossary nor select a domain: AutoTerm-SST selects the active resource from streaming evidence and injects at most 10 retrieved terms into each LLM prompt. This allows new domains to be added without retraining the translation model or collapsing all terminology resources into a single monolithic inventory.

AutoTerm-SST treats terminology control as an active-inventory and ranking problem. Its motivation is not that million-term retrieval is computationally infeasible, but that modular domain resources are easier to maintain, update, attribute, and keep relevant under a fixed prompt-evidence budget. Automatic mode starts from a deployment-configured default slice, and a training-free router switches among domain slices when streaming evidence is strong. Broad open memory is used only as an optional low-confidence fallback pool.

The result is an interactive streaming SST workbench rather than a new translation model. The framework handles REST/WebSocket transport, session lifecycle, and agent routing; agents handle model inference, prompting, batching, and retrieval. The live agent uses Qwen3-Omni with vLLM continuous batching (Kwon et al., 2023), and the web

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and Electron interfaces expose retrieved terms, active glossary status, router decisions, and latency alongside translations.

Our contributions are:

- We build AutoTerm-SST, a pluggable streaming SST demo with a live evidence panel for retrieved terminology, active glossary state, router decisions, and latency.
- We introduce zero session-time terminology setup: users neither upload a glossary nor select a domain, while a training-free router selects among prebuilt domain slices and sends at most 10 score-filtered candidates to the prompt.
- We evaluate terminology recall, retrieval latency and scale up to million-term memories, mixed-domain routing, and 32-session concurrent streaming.

2 Related Work

Streaming speech translation. SST systems must translate partial speech while balancing quality and latency. Prefix-to-prefix policies such as wait-k make this trade-off explicit (Ma et al., 2019), attention-based policies refine read/write decisions (Papi et al., 2023, 2024), SimulEval standardizes evaluation (Ma et al., 2020), and large-scale streaming models target direct simultaneous translation (Seamless Communication, 2023; Zhang et al., 2024). Speech language models make this setting more flexible (Qwen Team, 2025; Ouyang et al., 2025), but live deployment still requires session management, incremental transport, and latency-aware prompting. AutoTerm-SST focuses on adapting terminology during a stream; its contribution is a system demonstration rather than a new model architecture.

Terminology-aware translation. Terminology control has long been studied in machine translation, usually through user-provided dictionaries, constrained decoding, or training-time exposure to terminology constraints (Hokamp and Liu, 2017; Dinu et al., 2019); terms and named entities are also a known weakness of speech translation (Gaido et al., 2021). Retrieval-augmented generation offers another way to expose non-parametric memory at generation time (Lewis et al., 2020; Khandelwal et al., 2021). Many of these workflows assume that the domain or terminology is known before translation begins. RASST retrieves from a glossary fixed before streaming (Luo et al., 2026); AutoTerm-SST selects the active glossary online from accumulated streaming evidence while keeping the retriever and translation model unchanged.

LLM serving and demo systems. Modern LLM serving systems provide continuous batching and efficient generation APIs (Kwon et al., 2023; Zheng et al., 2024), but a streaming speech demo also needs microphone/file input, WebSocket transport, session cleanup, live evidence rendering, and mock-mode testing. Translation Canvas (Dandekar et al., 2024) targets offline translation analysis; AutoTerm-SST provides a live serving layer while keeping model-specific code in swappable agents.

3 System Overview

AutoTerm-SST uses a thin framework with swappable translation agents, while its main system contribution is the adaptive terminology loop inside the live agent (Figure 1). The framework serves the web UI, accepts REST control requests, streams audio over WebSockets, tracks session lifecycle, and routes each session by `agent_type`; it does not own model prompts, retrieval, batching, KV cache, or terminology resources. For chunk t, the agent retrieves from the current slice before decoding. The resulting streaming evidence may update the slice for a later chunk, no earlier than $t + 1$, but never changes the prompt already used for chunk t.

Agent contract. Agents implement a small contract: `open_session`, `submit_audio`, `on_control`, `close_session`, `describe`, and `health`. The same UI and wire protocol support a deterministic mock agent for development, a legacy InfiniSST scheduler agent (Ouyang et al., 2025), or the AutoTerm-SST live agent built on RASST/Qwen3-Omni (Luo et al., 2026). The boundary is backend-agnostic: the live demo uses in-process vLLM (Kwon et al., 2023), and the same protocol can serve external SGLang-compatible backends (Zheng et al., 2024) without changing the terminology memory, router, or UI.

Streaming agent. For each audio chunk, the OmniAgent runs MaxSim terminology retrieval over the active slice and collects routing-only domain-probe evidence. Before decoding, the hybrid router combines current speech/retrieval evidence with target-text history from earlier chunks; an accepted

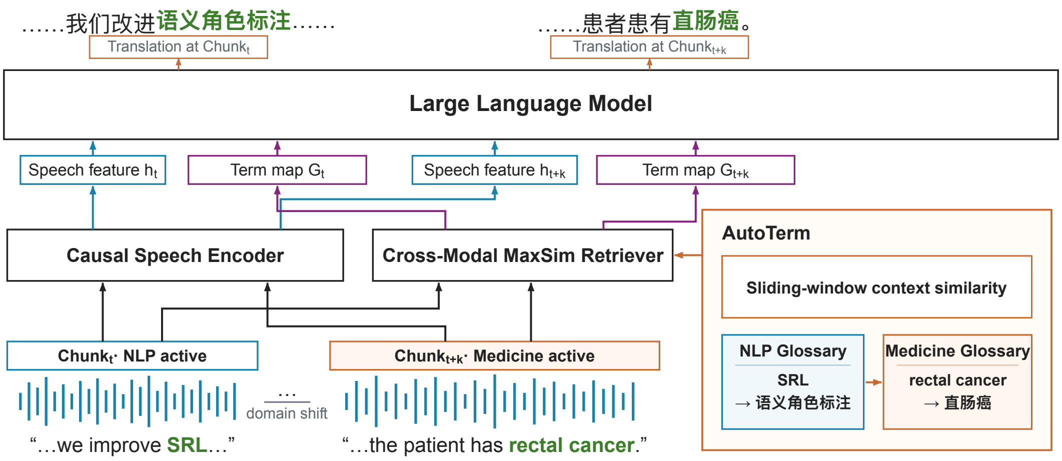

Figure 1: AutoTerm-SST on a stream that moves from NLP to medicine. Each speech chunk is encoded by the RASST backbone, and at most 10 bilingual term pairs retrieved from the *active slice* are injected into the Qwen3-Omni prompt. The router switches the active slice only after accumulated evidence confirms the domain shift, so *SRL* is served from the NLP slice early and *rectal cancer* from the medicine slice later.

switch affects only later retrieval. The agent filters and reranks candidates, caps prompt references at 10, and lets the PromptBuilder format the `term_map` before calling the in-process vLLM backend. After decoding, y_t is appended to the target history for a later routing tick. The agent emits partial translations and structured metadata, including retrieved terms, retrieval latency, active slices, prompt-reference counts, and router decisions.

Evidence protocol. The default WebSocket output remains plain text for backward compatibility. A JSON mode returns `{type, text, meta}`, where `meta.references` drives the live evidence panel and `meta.topic` and `meta.topic_router` describe the adaptive glossary state. This evidence protocol lets users see which terms were retrieved and whether the router stayed, switched, or deferred.

4 Adaptive Terminology Memory

AutoTerm-SST separates terminology inventory from the prompt interface. Its *terminology memory* is a collection of prebuilt inventories derived from Wikidata/Wikipedia and curated glossaries. At any time, the *active slice* is the compact domain inventory queried for each chunk; a broad *open memory* can be consulted as an optional fallback. The online system retains at most 10 score-filtered references for each prompt.

Open memory snapshots. Each snapshot maps a preset identifier to term files, metadata, and a MaxSim index. The runtime never stores large indexes in the code repository; instead, manifests point to a runtime root on the demo host. The released configuration provides NLP and medicine slices. A curated 238-term ACL glossary is used only as an evaluation oracle; the 100k open memory is an optional fallback, and the 1M memory is used for scale diagnostics.

Hybrid active glossary router. The default live router is `hybrid_window_topic`. It is not a trained topic classifier and does not run a separate ASR path or source-transcript classifier in production. Instead, it combines signals available within the streaming agent: windows of recent generated target text, routing-only speech-window domain probes, weak speech-centroid evidence, and retrieved-reference metadata:

$$\begin{aligned} score(d) =& \lambda_t \,\mathrm{topic}_d(y_{\mathrm{recent}}) \\ &+ \lambda_p \,\mathrm{probe}_d(x_{\mathrm{speech}}) \\ &+ \lambda_c \,\cos(\mathrm{EMA}(q_{\mathrm{speech}}), c_d) \\ &+ \lambda_m \,\mathrm{vote}_d(\mathrm{references}), \end{aligned} \tag{1}$$

where y_{recent} is the recent generated translation window, x_{speech} is the recent speech window used for a routing-only domain probe, q_{speech} is the pooled speech query embedding, c_d is a domain centroid built offline from a slice's MaxSim text embeddings, and reference votes come from metadata such as source preset and domain. Here, topic_d is a weighted keyword/regex score over the generated target window, while probe_d aggregates top-k MaxSim scores between the speech window and each candidate domain index. The weights and switch thresholds are hand-set: "training-free" refers only to the router and does not imply that the base model or retriever is untrained. The default weights are $\lambda_t = 0.60$, $\lambda_p = 0.25$, $\lambda_c = 0.10$, and $\lambda_m = 0.05$; they are renormalized over the signals available in a window, so routing already works before the first target text is generated. The production schedule evaluates switch decisions every 10s after a 20s warmup and caches probe scores between refreshes (Appendix A). The older `embedding_refs` router remains available as a compatibility policy.

Conservative switching. The router is a policy for selecting active domain slices, not an open-world topic detector. It switches only after warmup, minimum update interval, confidence threshold, top-vs-runner-up margin, repeated consistent windows, domain-probe guards, and target index readiness. A switch supported by generated target text requires multiple consecutive consistent windows; audio/probe-only switching uses stricter raw-probe floors (Appendix A). Generated target text without positive topic evidence receives no topic weight, so it cannot dilute a strong probe or turn a weak probe into a high-confidence switch. Cold index loads are non-blocking: the agent preloads a target index in the background and activates it only after the retrieval plugin reports that the index is ready.

Capped prompt candidates. Automatic mode retrieves and score-filters up to 10 candidates from the active slice. Under low-confidence fallback conditions, it may query `open_wiki_100k` separately, then merge, deduplicate, and rerank the two result sets. The prompt receives at most $\texttt{PROMPT\_K} = 10$ references, with no filler terms when fewer survive. Slice size therefore trades coverage against ranking difficulty within the top-10 prompt cap. The controlled comparison uses 10k inventories as an evaluation choice, not a system limit; broader inventories remain available as fallback pools and scale diagnostics (Appendix B). The live evidence metadata records active slices, candidate-pool size, prompt-reference counts, fallback events, and retrieval-quality metrics.

5 Demo Interface

The demo is designed for three audiences: researchers inspecting terminology errors in streaming SST, lecture or conference participants who do not want to prepare glossaries, and developers testing new speech translation agents behind a stable protocol. The same interface can run in mock mode without a GPU or in live GPU mode with Qwen3-Omni and MaxSim retrieval.

Live controls. Users choose a model agent, language pair, latency multiplier, and terminology mode. In `auto_working` mode, the session starts from a configured default slice without requiring the user to identify the domain, and it may switch to another domain slice; every chunk uses the same top-10 prompt cap. The hosted demo serves a lightly curated variant of the two slices (generic single-word entries removed, session proper nouns added); all evaluation numbers use the unmodified union inventories of Section 6. Manual presets and ad-hoc term overrides remain available as advanced options; they affect the prompt but do not bias the router.

Evidence panel. For each partial translation, the UI renders the latest non-empty retrieved references as term-translation pairs with scores and source labels. A status line reports active memory backend, term count, and retrieval/generation latency; the adaptive row adds mode, inferred topic, confidence, active glossary, switch count, and router action. This makes terminology behavior visible rather than hidden inside the model prompt.

Packaging. The repository includes a mock-friendly launcher for UI/protocol development and a GPU launcher for the live AutoTerm-SST agent built on RASST. Mock mode exercises the same REST/WebSocket flow and JSON evidence protocol, enabling reviewers to test the demo without model weights or GPUs. The live demo additionally uses a resident vLLM engine, a dedicated retriever GPU, and an optional public tunnel.

Availability and licensing. The demo code is released under the MIT license on the GitHub repository. A source checkout is the downloadable package; in mock mode, it starts the same UI and protocol without GPUs. Live Qwen3-Omni mode requires A6000-class GPUs and the external RASST retriever checkpoint; model weights and Wikidata/Wikipedia-derived terminology resources retain their upstream licenses. The hosted demo is

Figure 2: Web UI during a live En→Zh session, moments after the automatic switch to the medicine slice. The translucent panel renders the JSON evidence stream over the settings card: retrieved term pairs with retrieval scores and their source slice (`auto:medicine_core_10k`), per-chunk retrieval latency, and the streaming translation with retrieved terminology hits highlighted inline. The latency multiplier was changed live mid-session (status chip).

reachable through a stable entry page.[1]

6 Evaluation

Setup. We evaluate terminology memory and routing in the live AutoTerm-SST system on A6000 GPUs. Its live agent runs Qwen3-Omni with in-process vLLM tensor parallelism over two GPUs and the RASST MaxSim retriever on a third GPU. Audio is from prepared ACL 60/60 English-to-Chinese and medicine segments. Unless otherwise stated, results use the JSON WebSocket protocol, which exposes translated text, retrieved references, active glossary metadata, and per-chunk retrieval latency. A latency multiplier of 2 corresponds to 1.92s input chunks. We report three complementary terminology-memory evaluations — a controlled mixed-domain terminology comparison, a warm-retrieval scale sweep, and a mixed-domain routing probe — alongside single-session real-time serving.

Metrics. For terminology quality, term accuracy is the fraction of annotated target-term occurrences translated. Each occurrence is anchored to source-audio seconds by forced alignment and counts as a hit only when a distinct target variant appears in the output within a $[-2, +30]$ s window, so repeated occurrences are scored independently rather than crediting a talk for one appearance (Appendix C). The controlled comparison contains 1,462 technical occurrences: 780 from ACL and 682 from medicine, the latter spanning 196 block-

[1] Live demo: https://luojiaxuan.github.io/autoterm-sst/; screencast video: https://github.com/luojiaxuan/autoterm-sst/blob/main/docs/demo_screencast.mp4.

Setting	Session setup	Term acc.	ACL	Med.	BLEU
No memory	None	0.670	0.767	0.559	58.84
Fixed NLP	Select NLP	0.679	0.797	0.544	59.18
Fixed medicine	Select med.	0.805	0.764	0.852	**59.63**
Automatic	None	**0.874**	**0.890**	**0.856**	58.10

Table 1: Term accuracy on the ten-talk alternating ACL/medicine stream (16,848s, one live session; 1,462 gold occurrences: 780 ACL + 682 medicine). ACL/Med. = term accuracy within each domain's talks. Session setup = what the user does at session time. Every inventory is a 10k-term union that fully covers its own domain's gold terms.

local medicine term types. Masked BLEU deletes annotated target-term strings from both hypothesis and reference before scoring, and is used only as a non-terminology sanity check. For runtime, we report prompt references per partial chunk and retrieval-latency percentiles. For routing, we report switch latency, wrong switches, and steady-state active-domain accuracy after transition grace windows. These metrics do not replace token-delay measures such as StreamLAAL (Papi et al., 2024).

Controlled terminology comparison. Table 1 compares equal-sized domain inventories that fully cover the gold terms in their matching domain. It therefore tests whether routing can compose domain-specific resources across a stream, rather than evaluating open-world terminology mining. Automatic mode reaches 0.874 combined term accuracy, exceeding both fixed inventories overall and each domain expert within its own domain (ACL 0.890 vs. 0.797; medicine 0.856 vs. 0.852), at a 1–1.5 BLEU cost from switch windows. Appendix C adds raw-gold, type-level, and masked-BLEU detail plus a three-block probe. A five-talk sweep

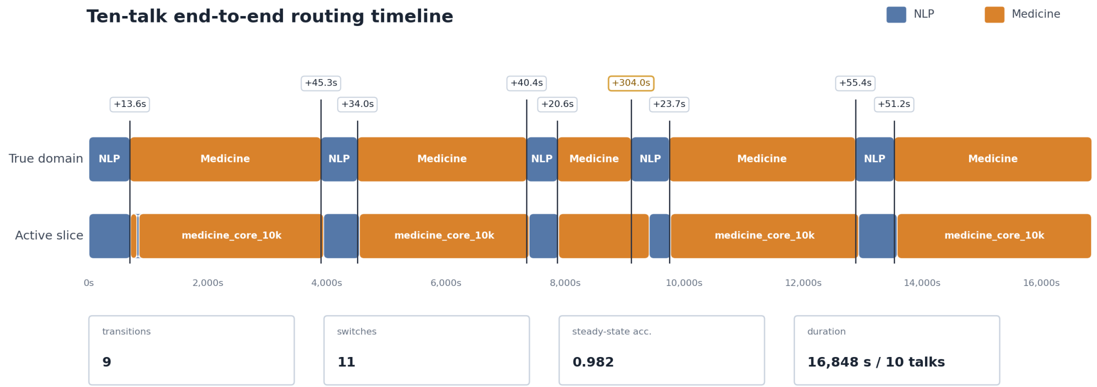

Figure 3: Active-slice routing over the ten-talk run of Table 1: nine domain transitions, eleven switches. Vertical lines mark true domain boundaries; chips give the switch delay after each boundary; the amber switch crosses a generic talk opening.

shows stable term accuracy (0.965–0.979) while retrieval precision falls 0.48 → 0.14 as distractors grow (Appendix B).

Scale and prompt noise. Warm MaxSim retrieval stays near 60–105 ms (p50–p95) from curated glossaries to million-term memories; the scaling costs are cold index loading and reference noise rather than per-chunk retrieval, which is why the active slice stays compact and broad memories serve as fallback pools (Table 3, Appendix B).

Mixed-domain routing. Figure 3 traces active-slice routing across the ten-talk run of Table 1: nine correct-direction transitions, one delayed switch over a generic talk opening, and one self-correcting 35s transient, with the top-10 prompt invariant held throughout (Appendix C). A shorter four-block probe (480s) behaves the same way: three switches 17.3–37.4s after the boundaries, zero wrong switches, 99.5% steady-state accuracy. A medicine-only session started from the default NLP slice recovers by switching to the medicine slice 59.5s into the session with zero wrong switches, so the initial slice is a warm start rather than a hidden oracle. The speech probe alone is noisy (0.52 top-1 accuracy), so generated-target evidence leads routing and the probe acts as a guard. Two 640-window state-machine proxy tests, using reference-text windows and controlled probe evidence, cover 16 alternating/random transitions with zero wrong switches. Inverted- and contested-probe controls expose the expected wrong-switch and delayed-switch failure modes. These proxies are separate from the preceding end-to-end audio runs.

Serving. A single live session sustains the 1.92s real-time input rate (median first-output latency 0.35s; warm per-chunk retrieval 60–105 ms). Multiple sessions are served concurrently through vLLM's batched inference, used off the shelf.

7 Limitations and Conclusion

AutoTerm-SST is a live terminology-aware SST system, not a new foundation model or topic classifier. The live demo serves En→Zh/Ja/De; routing depends partly on small, hand-written target-language keyword lists, so unsupported languages or domains may switch more slowly (Appendix D). Large open memories can expand coverage but reduce retrieval precision under the 10-reference cap, so the 100k memory is queried only through a guarded fallback. The hybrid router is intentionally conservative and may retain the initial slice when evidence is ambiguous. Our evaluations are single-run system probes, and the matched-union comparison uses full-coverage benchmark inventories; neither establishes open-world or corpus-level significance. The retrieval-latency sweep excludes the routing-only domain probes, whose cost the runtime records separately. Finally, the real backend needs high-end NVIDIA GPUs and external model/retriever artifacts; mock mode covers UI and protocol testing without them.

We presented AutoTerm-SST, a terminology-memory workbench that requires no per-session glossary upload or domain choice. It keeps broad resources off the default retrieval path, selects active slices from streaming evidence, injects at most 10 score-filtered references per prompt, and surfaces live terminology and router evidence.

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A Router Configuration and Guards

Table 2 lists the default production configuration of the hybrid active-glossary router as released in `configs/autoterm_slices.yaml`; deployment-specific configurations can override individual values explicitly. Guards are evaluated in order: warmup, update interval, target-equals-active, index readiness, probe floors, confidence, top-vs-runner-up margin, target-vs-current margin, post-switch cooldown, and consecutive-window consistency. Uncertain windows keep the current slice. The released configuration uses two consistent generated-target windows and a 0.30 target-vs-current margin; the two 640-window proxy tests in Section 6 used three generated-target windows.

The weights are hand-set on the proxy schedules, not fitted to the audio evaluation runs. A proxy weight sweep (λ_t from 0.25 to 0.80, including dropping the centroid and metadata terms) leaves switching unchanged: every mixed-signal configuration completes all 16 alternating/random transitions with zero wrong switches, at a latency fixed by the three-window consistency guard. Setting λ_t=1 (no probe channel) blocks most switches (1/9 and 2/7), so the speech probe is a required corroborating guard even though it is too noisy to lead (Section 6).

Parameter	Default
Signal weights (renormalized per window)	
generated-target topic λ_t	0.60
speech-window domain probe λ_p	0.25
speech centroid λ_c	0.10
reference metadata λ_m	0.05
Switch thresholds	
activation confidence	0.60
top-vs-runner-up margin	0.15
target-vs-current margin	0.30
consistent windows (text / gen. target / audio)	2 / 2 / 3
Audio-probe floors / schedule / retrieval	
probe top score / margin / positive domains	0.50 / 0.08 / 2
warmup / update / cooldown / staleness	20s / 10s / 30s / 120s
cap `PROMPT_K` / score threshold	10 / 0.78

Table 2: Released hybrid-router defaults. Signal weights are renormalized over the signals present in each window; the three consistent-window counts apply per evidence path (manifest text / generated target / audio-only).

B Glossary-Scale Streaming Sweep

Table 4 reports the five-talk streaming sweep referenced in Section 6: five concatenated ACL 60-60 talks (3,442s of audio) streamed through the live JSON WebSocket at latency multiplier 2, scored against a curated 142-term technical gold set. Each inventory always contains the full 238-entry curated ACL glossary; larger rows add Wikidata-mined AI/CS candidates as distractors. These are manual-preset runs isolating inventory scale (not `auto_working` runs). Term accuracy, BLEU, and masked BLEU stay in a narrow band up to 100k terms while retrieval precision collapses, motivating compact routed active slices.

Memory	Terms	p50/p95 ms	Refs/chunk
Curated	238	61.3 / 105.3	2.33
Domain	10k	78.5 / 82.1	1.80
Open	100k	64.1 / 82.6	4.89
Stress	1M	78 / 95	9.62

Table 3: Warm per-chunk MaxSim retrieval latency (p50/p95) and surviving prompt references per chunk (Refs) by memory size. The cost that scales is cold index loading (0.5s at 238 terms, 6.8s at 100k, over 30s at 1M), which AutoTerm-SST does off the streaming path.

Inventory	Terms	Term acc.	BLEU	M-BLEU	Gold@10	Prec@10	Refs
Curated ACL	238	0.972	58.27	53.10	0.972	0.480	1.53
+AI transl. 10k	10,238	0.965	58.72	53.35	0.972	0.234	3.09
+AI broad 10k	10,238	0.972	58.38	52.75	0.979	0.367	2.00
+AI broad 50k	50,238	0.979	58.65	53.24	0.979	0.196	3.64
+AI broad 100k	100,238	0.965	58.83	53.32	0.979	0.137	5.03

Table 4: Five-talk streaming sweep (latency multiplier 2) against the 142-term technical gold set. Gold@10 = fraction of gold terms surfaced within the top-10 prompt cap; Prec@10 = retrieval precision at 10; Refs = surviving prompt references per chunk; M-BLEU = BLEU after deleting gold term strings from hypothesis and reference. At latency multiplier 1 the pattern holds (term accuracy 0.944–0.958; precision 0.479 → 0.133).

C Mixed-Domain Terminology under Union Inventories

A glossary-channel comparison is only meaningful when the gold terms are inside the retrieval inventory: the broad wiki slices cover 29/142 of the ACL technical gold and 1/54 of the `medicine_606` gold, so their term accuracy mostly measures base-model retention. We therefore evaluate with *benchmark-aligned union inventories* at a fixed 10k inventory size: the byte-identical gold glossary (238 ACL / 212 hard-medicine terms) plus seeded wiki fillers, mirroring the recipe of Appendix B (gold coverage 142/142 and 54/54).

Occurrences are anchored to source-audio seconds by Montreal Forced Aligner word alignments (ACL, through the played segment offsets) or oracle span times (medicine); a hit needs a distinct target variant within a $[-2, +30]$ s window, matched one-to-one. Alignments ship with the code (`eval/streaming_sst/mfa_alignments`) and the lead is stable for 15–45 s windows.

Table 5 gives the full detail for the ten-talk run of Table 1 (five ACL and five medicine talks alternating, 16,848s streamed live in real time through one session): raw-gold and type-level term accuracy plus masked BLEU. Automatic mode requires no session-time glossary input or domain choice but relies on inventories built offline. All nine domain transitions switch in the correct direction with one transient extra switch (steady-state active-domain accuracy 0.982); eight of nine switches land within 13.6–55.4s, and one delayed medicine-to-ACL switch takes 304s over a generic talk opening. A truncated three-block probe (ACL → medicine → ACL, 600s per block) shows the same ordering — Automatic 0.908/0.888 combined versus 0.888/0.874 (fixed NLP) and 0.738/0.727 (fixed medicine), with switches 37.4s/25.0s after the boundaries — and a gold-only 212-term medicine inventory without fillers reaches just 0.738, confirming that compact curated slices need distractor mass to be a realistic comparison. This probe is ACL-heavy (367 of 412 technical occurrences), so the fixed-NLP inventory is a close runner-up.

Setting	$\mathrm{Acc}_{\mathrm{tech}}$	$\mathrm{Acc}_{\mathrm{raw}}$	ACL	Med.	Med. types	BLEU	M-BLEU
None	0.670	0.706	0.767	0.559	92/196	58.84	57.46
Fixed NLP	0.679	0.746	0.797	0.544	86/196	59.18	57.85
Fixed medicine	0.805	0.784	0.764	0.852	168/196	**59.63**	57.53
Automatic	**0.874**	**0.868**	**0.890**	**0.856**	**170/196**	58.10	56.22

Table 5: Full detail for the ten-talk run. $\mathrm{Acc}_{\mathrm{tech}}$/$\mathrm{Acc}_{\mathrm{raw}}$ = term accuracy under the curated technical gold (1,462 occurrences) and the raw annotation gold (2,551); ACL/Med. = per-domain accuracy under the technical gold; Med. types = medicine term types hit at least once (of 196); M-BLEU as in Table 4.

D Multilingual Terminology (En → Ja/De)

The terminology resources provide three target-language translations: the gold glossaries carry zh/ja/de values, a single MaxSim index serves every target language (translations are resolved from the term list at prompt time), and the language pair selects the per-language demo model. Table 6 repeats the mixed full-length ACL → medicine → ACL comparison for En→Ja and En→De. Automatic routing, with no session-time glossary input, has the best combined term accuracy in both languages under both gold denominators, with zero wrong switches and steady-state active-domain accuracy 1.0. Both languages route ACL better than the fixed-NLP inventory and reach medicine coverage without a medicine glossary, so the combined lead comes from balanced cross-domain coverage rather than winning either domain outright. With the released ja/de topic keywords, switches land within 29–52s (ja 28.9/33.5s; de 52.0/33.5s; zh 37.4/25.0s). An ablation without language-tuned keywords degrades first-switch latency to 115.3s (ja) and 875.6s (de), when routing leans on the slower probe channel. These delays coincide with lower medicine coverage and BLEU while switch directions stay correct; the keyword lists are small, high-precision regex sets per domain. German term matching is casefolded with light stem tolerance (exact matching undercounts under inflection); short variants keep strict word boundaries. The medicine references differ in provenance: German uses human reference translations, whereas Chinese and Japanese medicine references are model-generated (the ACL segments use the shared-task human references in all three). Under the human-referenced German setting, automatic routing still leads on the raw gold and stays within 1.1 points of the in-domain expert on the technical gold without any session-time setup.

	Setting	$\mathrm{Acc}_{\mathrm{tech}}$	$\mathrm{Acc}_{\mathrm{raw}}$	ACL	Med.	Med. types	BLEU	M-BLEU
Ja	None	0.635	0.669	0.730	0.578	25/53	43.55	41.60
	Fixed NLP	0.643	0.703	0.900	0.488	21/53	44.19	42.21
	Fixed medicine	0.827	0.811	0.720	**0.892**	**48/53**	**44.90**	**42.83**
	Automatic	**0.857**	**0.860**	**0.930**	0.813	43/53	39.68	37.67
De	None	0.653	0.669	0.740	0.604	24/49	37.52	37.47
	Fixed NLP	0.667	0.694	0.840	0.571	23/49	37.21	36.91
	Fixed medicine	**0.812**	0.790	0.670	**0.890**	**43/49**	**37.63**	**37.18**
	Automatic	0.801	**0.795**	**0.880**	0.758	38/49	35.90	35.37

Table 6: En→Ja/De mixed-domain comparison; columns as in Table 5 (ja: 266/350 technical/raw gold occurrences, 53 medicine types; de: 282/366, 49). All runs use the released configuration with the ja/de topic keywords.